Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Mid-Semester Test (EC-2 Make-up)

Course No. : AIMLCZG512 **Course Title**: Deep Reinforcement Learning
Nature of Exam : Closed Book **Weightage** : 30% **No. of Pages** = 2 ; **No. Of Questions =** 5;
Duration : 2 Hours; **Date of Exam**: 28-07-2024 (EN)

Note to Students:
1. Answer all the questions.
2. Write your name and sign at the end of all the pages.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

We have considered your correct and alternative solutions even If it does not match the scheme given in this scheme

Q1) [Write the answer in an A4 Sheet, Scan & Upload]

(a) In what ways a Unsupervised Learning and Reinforment Learning are different? [2.0 M]

Answer Key: Two aspects Unsupervised learning and reinforcement learning are different are as below.

(1) On the Objective: Unsupervised learning focuses on finding hidden structures in unlabeled data, such as clustering or dimensionality reduction. In contrast, reinforcement learning (RL) is concerned with maximizing a reward signal through interactions with an environment learning to make sequences of decisions. [1 marks]

(2) Learning Process: While unsupervised learning seeks patterns without specific feedback, RL involves an agent learning from the consequences of its actions, adjusting behavior to achieve the highest possible reward, rather than just uncovering underlying data structures. [1 marks]

(b) How are model based and model free reinforcement algorithms differ? What are scenarios when these approaches are appropriate? [scenarios in not more than three statements] [2.0 M]

Answer Key:

(1) In model-based RL, the agent uses or learns a model of the environment. This model allows the agent to plan its actions by simulating different sequences of actions and choosing the one that maximizes long-term rewards. Classical dynamic programming (DP) methods are model-based. Situations where model dynamics are available , model-based methods are suitable. [1 marks]

(2) Model-free RL does not use an environment model. Instead, the agent directly learns from interactions with the environment, adjusting its behavior based on observed rewards and state transitions. Monte Carlo methods are examples. Situations where adequate data in the form of episodes are available, model free methods are suitable. [1 marks]

The scenario could be the specific ones that the students are expected to give.

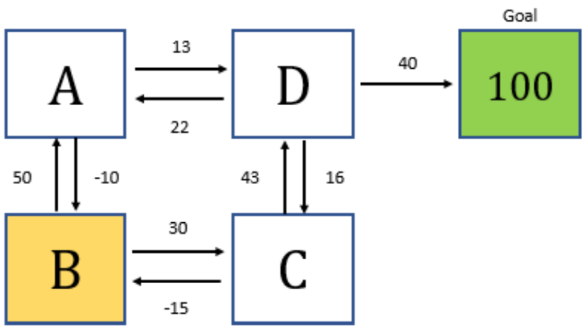
(c) Write any one use-case (application) from your workplace which can be modeled as a RL problem suitable to be solved by dynamic programming approach. Your answer should have all the elements that identify the use case sitable to be solved by dynamic programming approach. [vague description of application will be penalized]. [2.0 M]

Scenario where the details of the model can be explicit - transition model, reward model - are accepted.

[2.0 + 2.0 +2.0 = 6.0 Marks]

Q2) [Write the answer in A4 Sheet, Scan & Upload]

(a) In the image below is a representation of the game that you are about to play. There are 5 states: A, B, C, D, and the goal state. The goal state, when reached, gives 100 points as reward. Treat this as the value of being in the Goal State. The immediate rewards while transitioning is mentioned in the edges. Assume the values of states are initialized to 0. Assume that the each outgoing actions from the states are equiprobable. Assume we are using asynchronous value iteration with a decay (γ) of 0.9. Show the value of each state after one iteration, if the states are considered in the order D, A, C, B. What will be the policy at state B after this iteration.



[6 Marks]

Answerk Key:

Given:

States: A, B, C, D, and the Goal state.
Initial Values: V(A)=V(B)=V(C)=V(D)=0
Goal state reward: 100.
Discount factor γ=0.9.
The states are updated in the order: D, A, C, B.

From State D: [1.25 Mark]

Possible Actions from D:

Move to A : R = 22 + γ.V(A) = 22+ 0 = 22
Move to C : R = 16 + γ.V(C) = 22+ 0 = 16
Move to Goal : R = 40 + γ.100 = 44+ 90 = 130

Value for State D (equiprobable actions):

V(D) = ⅓(22+16+130) = 168/3 = 56

From State A: [1.25 Mark]

Possible Actions from A:

Move to B : R = -10 + γ.V(B) = -10+ 0 = -10
Move to D : R = 13 + γ.V(D) = 13+ 0.9*56 = 63.4

Value for State A:

V(A) = ½(-10+63.4)=26.7

From State C: [1.25 Mark]

Possible Actions from C:

Move to B : R = -15 + γ.V(B) = -15+ 0 = -15

Move to D : R = 43 + γ.V(D) = 43+ 0.9*56 = 93.4

Value for State C:

V(C) = ½(-15+93.4)=39.2

From State B: [1.25 Mark]

Possible Actions from B:

Move to A : R = 50 + γ.V(A) = 50+ 0.9*26.7 =74.03

Move to C : R = 30 + γ.V(D) = 30+ 0.9*39.2 = 65.28

Value for State B:

V(B) = ½(74+65.28)=69.65

Policy at State B: [1 Mark]

The possible values after one iteration:

Move to A: 74.03

Move to C: 65.28

Since 74.03 > 65.28, the optimal policy at state B is to **move to state A**.

Q3) [Write the answer in A4 Sheet, Scan & Upload]

(a) In the gridworld example, rewards are positive for goals, negative for running into the edge of the world, and zero the rest of the time. Are the signs of these rewards important, or only the intervals between them? Explain. [2.0 M].

Solution

Lets assume the reward for positive goal is +x and negative goal is -y; Adding y+c to the rewards of both positive and negative shifts the rewards to be positive for both goals. When we shift the regards by a constant C, the relative values shifts by a constant as demonstrated below:

$$\begin{aligned} v'_\pi(s) &= E_\pi\{\sum_{k=0}^{\infty}\gamma^k(R_{t+k+1}+C)|s_t=s\} \\ &= E_\pi\{\sum_{k=0}^{\infty}\gamma^k R_{t+k+1} + C\sum_{k=0}^{\infty}\gamma^k|s_t=s\} \\ &= v_\pi(s) + \frac{C}{1-\gamma}. \end{aligned}$$

This shift applies to all the values if the rewards are shifted by a constant.

For evaluation, following are expected from students:

(1) Stating only intervals are important - 1 M

(2) Attempt to demonstrate (even if it is incorrect) the correspondence between values when the rewards are shifted by C. - 1 M

(b) Consider a k-armed bandit problem with k = 5 actions, denoted 1, 2, 3, 4 and 5. Consider applying to this problem a bandit algorithm using **ε**-greedy action selection, sample-average action-value estimates, and initial estimates of Q1(a) = 4, for all a. Suppose the initial sequence of actions and rewards is A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5. On some of these time steps the ε-case may have occurred, causing an action to be selected at random.

(i) On which time steps did this definitely occur? Why? [2 M].

(ii) On which time steps could this possibly have occurred? Why? [2 M].

[2.0 + 4 = 6 Marks]

Solution Scheme:

Given

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

Q1(1) = 4 ; Q1(2) = 4; Q1(3) =4; Q1(4) = 4; Q1(5) =4

Q value for all the actions are the same and A1 has been selected.

⇒ A1 is Possibly greedy (tie-braking by random) or epsilon-greedy

Now;

Q1(1) = 3 ; Q1(2) = 4; Q1(3) =4; Q1(4) = 4; Q1(5) =4

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

Q value for all the actions except A1 are the same and A2 has been selected.

⇒ A2 is Possibly greedy (tie-braking by random) or epsilon-greedy

Now

Q1(1) = 3 ; Q1(2) = 3; Q1(3) =4; Q1(4) = 4; Q1(5) =4

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

Q value for all the actions except A1 , and A2 are the same and A5 has been selected.

⇒ A5 is Possibly greedy (tie-braking by random) or epsilon-greedy

Now

Q1(1) = 3 ; Q1(2) = 3; Q1(3) =4; Q1(4) = 4; Q1(5) =2.5

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

The highest values are for actions 3 and 4. If greedy / tie-braking by random, one of these actions should have been selected. However, the action selected is 2 and hence the epsilon case is applied.

Now

Q1(1) = 3 ; Q1(2) = 2.33; Q1(3) =4; Q1(4) = 4; Q1(5) =2.5

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

The highest values are for actions 3 and 4.

The highest values are for actions 3 and 4. Action 4 has been taken. This implies, either greedy / tie-braking by random / epsilon greedy could have be the case.

Now

Q1(1) = 3 ; Q1(2) = 2.33; Q1(3) =4; Q1(4) = 2.5; Q1(5) =2.5

A_1 = 1, R_1 = 2, A_2 = 2, R_2 = 2, A_3 = 5, R_3 = 1, A_4 = 2, R_4 = 1, A_5 = 4, R_5 = 1, A_5 = 3, R_5 = 5.

The highest values are for action 3, and action three has been selected.
This can be due to greedy / e-epsilon greedy.

Q4) [Write the answer in A4 Sheet, Scan & Upload]

(a) Devise one example task of your own that fit into the MDP framework, identifying for each its states, actions, and rewards. Make the examples as different from each other as possible. [1 M] .

SOLUTION:
The example could be like any of the following 2 examples [Not limited to this].

Example 1: Investment Portfolio Management

1. Scenario: [0.25 Marks]

An investment manager is tasked with optimizing a portfolio of stocks, bonds, and other assets over time, balancing the risk and return to maximize the portfolio's value.

2. MDP Components:

- **States (S): [0.25 Marks]**
  - **Current Portfolio Allocation, Market Conditions, Interest Rates, Inflation Rate**
- **Actions (A): [0.25 Marks]**
  - **Buy Stocks/Bonds, Sell Stocks/Bonds, Hold Cash, Rebalance Portfolio**
- **Rewards (R): [0.25 Marks]**
  - **+X% for Portfolio Growth, -Y% for Portfolio Loss, -Z for High Risk, +W for Low Risk**

3. Objective:

Maximize the long-term growth of the portfolio while managing risk according to the investor's risk tolerance and market conditions.

Example 2: Personalized Fitness Coaching

1. Scenario: [0.25 Marks]

A personalized AI fitness coach is guiding a user through a workout plan aimed at improving overall fitness, including strength, flexibility, and endurance.

2. MDP Components:

- **States (S): [0.25 Marks]**
  - **Current Fitness Level, Current Fatigue Level, Previous Exercise Type, Time of Day**
- **Actions (A): [0.25 Marks]**
  - **Cardio Exercise, Strength Training, Flexibility Training, Rest**
- **Rewards (R): [0.25 Marks]**
  - **+10 for Progression, -5 for Overtraining, +5 for Consistency, -2 for Skipped Sessions**

3. Objective:

Maximize the user's fitness level and overall well-being by providing personalized workout recommendations while avoiding overtraining and ensuring consistency.

(b) Write a recursive expression that can be used to evaluate $q_\pi(s,a)$? What is the significance of the recursive nature of $q_\pi(s,a)$? [2 M].

SOLUTION

The function $q_\pi(s,a)$ represents the expected return (cumulative reward) of taking action a in state s and then following policy π thereafter. The recursive expression for $q_\pi(s,a)$ is: [0.5 Marks for the equation and 0.5 marks for explaining the equation].

$$q_\pi(s,a) = \sum_{s'} P(s'|s,a)\left[R(s,a,s') + \gamma \sum_{a'} \pi(a'|s') q_\pi(s',a')\right]$$

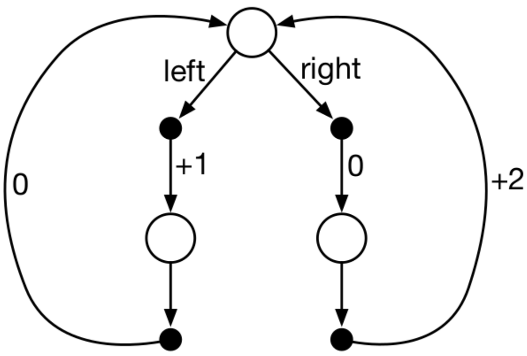
Where:

- s is the current state.
- a is the action taken in state s.
- s' is the next state after taking action a in state s.
- $P(s'|s,a)$ is the transition probability of moving from state s to state s' given action a.
- $R(s,a,s')$ is the reward received after transitioning from state s to state s' by taking action a.
- γ is the discount factor, which determines the importance of future rewards.
- $\pi(a'|s')$ is the probability of taking action a' in state s' under policy π.
- $q_\pi(s',a')$ is the expected return starting from state s', taking action a', and then following policy π.

SIGNIFICANCE OF THE RECURSIVE EQUATION: [1 Mark]

1. **Bellman Equation**
2. **Dynamic Programming**
3. **Exploration of Future States**
4. **Convergence**
5. **Policy Improvement**

(c) Consider the continuing MDP shown below.



The only decision to be made is that in the top state, where two actions are available, left and right. The numbers show the rewards that are received deterministically after each action. There are exactly two deterministic policies, $\boldsymbol{\pi}_{\text{left}}$ and $\boldsymbol{\pi}_{\text{right}}$.

(i) What policy is optimal if γ= 0.7? [1.5 M]. Explain

(ii) What policy is optimal if γ= 0.5 ?[1.5 M]. Explain

SOLUTION:

[For each sub section (i) and (ii) of the questions, value for $\boldsymbol{\pi}_{\text{left}}$ and $\boldsymbol{\pi}_{\text{right}}$ carries 0.5 marks each and conclusion carried 0.5 Marks]

Understanding the MDP:

- There is one decision state (top state) where the agent can choose between two actions: **left** or **right**.
- The rewards received after each action are:
  - **Left** action: $+1$ reward followed by a loop back to the top state.
  - **Right** action: $+2$ reward followed by a loop back to the top state.

Deterministic Policies:

- π_{left}: Always choose the **left** action.
- π_{right}: Always choose the **right** action.

Task:

Evaluate the optimal policy for two different discount factors: $\gamma = 0.7$ and $\gamma = 0.5$.

(i) Optimal Policy for $\gamma = 0.7$:

To find the optimal policy, we compare the expected return (value) for each action using the given γ.

- **Value for π_{left}:**
  - The expected return when choosing **left**:

$$V_{\pi_{\text{left}}} = 1 + \gamma V_{\pi_{\text{left}}}$$

$$V_{\pi_{\text{left}}} = 1 + 0.7 V_{\pi_{\text{left}}}$$

$$V_{\pi_{\text{left}}} - 0.7 V_{\pi_{\text{left}}} = 1$$

$$0.3 V_{\pi_{\text{left}}} = 1 \implies V_{\pi_{\text{left}}} = \frac{1}{0.3} = 3.33$$

- Value for π_{right}:
  - The expected return when choosing **right**:

$$V_{\pi_{\text{right}}} = 2 + \gamma V_{\pi_{\text{right}}}$$

$$V_{\pi_{\text{right}}} = 2 + 0.7 V_{\pi_{\text{right}}}$$

$$V_{\pi_{\text{right}}} - 0.7 V_{\pi_{\text{right}}} = 2$$

$$0.3 V_{\pi_{\text{right}}} = 2 \implies V_{\pi_{\text{right}}} = \frac{2}{0.3} = 6.67$$

Conclusion for $\gamma = 0.7$:

- $V_{\pi_{\text{right}}} = 6.67$ is greater than $V_{\pi_{\text{left}}} = 3.33$.
- Optimal Policy: π_{right}.

(ii) Optimal Policy for $\gamma = 0.5$:

Again, compare the expected returns for each action.

- Value for π_{left}:
  - The expected return when choosing **left**:

$$V_{\pi_{\text{left}}} = 1 + \gamma V_{\pi_{\text{left}}}$$

$$V_{\pi_{\text{left}}} = 1 + 0.5 V_{\pi_{\text{left}}}$$

$$V_{\pi_{\text{left}}} - 0.5 V_{\pi_{\text{left}}} = 1$$

$$0.5 V_{\pi_{\text{left}}} = 1 \implies V_{\pi_{\text{left}}} = \frac{1}{0.5} = 2$$

- Value for π_{right}:
  - The expected return when choosing **right**:

$$V_{\pi_{\text{right}}} = 2 + \gamma V_{\pi_{\text{right}}}$$

$$V_{\pi_{\text{right}}} = 2 + 0.5 V_{\pi_{\text{right}}}$$

$$V_{\pi_{\text{right}}} - 0.5 V_{\pi_{\text{right}}} = 2$$

$$0.5 V_{\pi_{\text{right}}} = 2 \implies V_{\pi_{\text{right}}} = \frac{2}{0.5} = 4$$

Conclusion for $\gamma = 0.5$:

- $V_{\pi_{\text{right}}} = 4$ is greater than $V_{\pi_{\text{left}}} = 2$.
- **Optimal Policy:** π_{right}.

[1.0 + 2.0+ 3.0 = 6 Marks]

Q5) [Write the answer in A4 Sheet, Scan & Upload]

Consider the following episodes as experienced by an agent following a policy π with states {A,B,C,T} in which T is terminal state and actions {x,y,z}. The episodes are written using the representation { state - action - reward- nextState … } order, and the number in red within the bracket is the immediate reward.

E1: C - x - (0) - A - x - (+10) - C - x - (+2) - B - y - (+12) - C - x - (+ 20) - T
E2: A - x - (+10) - C - x - (+2) - B - y - (+12) - B - y - (+12) - C - x - (+ 20) - T
E3: C - x - (+2) - B - y - (+12) - C - x - (+ 20) - T

Assuming the initial Qπ(s,a) are all 0's, compute the revised Qπ(s,a) using on-policy first-visit MC control algorithm for ε-soft policies for one iteration only [4 M]. Assume the γ = 0.9. Write down the revised policy. [2 M].

[6 Marks]

Answer Key:

Q (s,a) - Initialized.

A			B			C			T		
x	y	z	x	y	x	x	y	z	x	y	z
0	0	0	0	0	0	0	0	0	o	o	o

Note that each iteration in the algorithm is about one episode only. Hence, this solution will use only E1. E2 and E3 are not required.

From episode 1, we can compute the values of

(C,x), (A,x), (B,y)

Q(C,x) = 0 + (0.9 *10) + (0.9 *0.9 *2) + (0.9 * 0.9* 0.9 * 12) + (0.9 * 0.9* 0.9 * 0.9 * 20)
= 32.49
Q(A,x) = (10) + (0.9 *2) + (0.9* 0.9 * 12) + (0.9* 0.9 * 0.9 * 20)
=36.1
Q(B,y) = 12+ (0.9 *20)
= 30

The revised Q table is

A			B			C			T		
x	y	z	x	y	x	x	y	z	x	y	z
36.1	0	0	0	30	0	32.49	0	0	o	o	o

The question did not give a value for ε, yet requests for an ε-soft policy. ε-greedy is ε-soft. You can assume a convenience ε value. If the ε is assumed to be 0.3, then the ε-greedy policy looks as below.

π(. |A)			π(. |B)			π(. |C)			T		
x	y	z	x	y	x	x	y	z	x	y	z
0.8	0.1	0.1	0.1	0.8	0.1	0.8	0.1	0.1	o	o	o

Please note that the probability of greedy action is maximum for each state. You are expected to give a policy of this kind, and the values could reflect which actions are greedy.

IF you write the policy as below, it will not be accepted. As the algorithm does not produce a deterministic policy.

A	B	C
x	y	x

Marking scheme for the question will be

Computing values for (C,x), (A,x), (B,y), each gets 1.5 marks;
For the correct policy, you get 1.5 marks